

Spatial aggregation alters population-weighted estimates of InSAR observation support

Zixuan Liu¹ and Wei Xiong^{2,3*}

¹Detroit Green Technology Institute, Hubei University of Technology, Wuhan, 430068, China.

²School of Electrical and Electronic Engineering, Hubei University of Technology, Wuhan, 430068, China.

³Department of Computer Science and Engineering, University of South Carolina, Columbia, 29201, SC, USA.

*Corresponding author(s). E-mail(s): xw@mail.hbut.edu.cn;

Abstract

Population-weighted assessments of interferometric synthetic aperture radar (InSAR) observation support depend on how population is allocated within reporting units. Spatial aggregation can change these assessments even when the observations are unchanged. We isolate this effect by comparing allocation rules under a fixed geographic domain, support field and population total. In the Chao Phraya footprint, uniform allocation within 1-km cells increases the mean support deficit by 57.7–57.9% relative to native population-model allocation, equivalent to 7.75–7.78 percentage points of the domain total. Within-cell covariance between population and support explains this difference, which persists after excluding boundary cells. Crossed population-model and interferogram-sampling comparisons preserve its direction but change its magnitude. The shift is 0.108–0.117 percentage points of total population for the matched Bangkok final processing mask; California effects are also smaller. Controlled missingness shows that small errors where estimates remain available can coexist with substantial unestimated population allocation. The primary deficit averages unsupported model allocation over area and interferogram pairs; it does not identify subsidence exposure. Population-weighted reporting should therefore preserve the spatial association between population and observation support and report sensitivity to allocation and sampling.

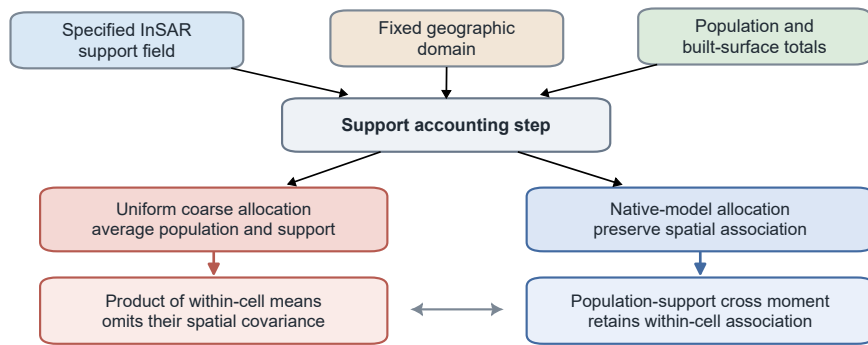
Keywords: InSAR, population allocation, observation support, coherence, spatial scale, reproducibility

1 Introduction

Satellite radar interferometry is now a routine source of surface-deformation information for Earth-system monitoring, urban hazard screening and infrastructure assessment. Synthetic Aperture Radar (SAR) missions such as Sentinel-1 provide repeated coherent observations that can be transformed into displacement products [1]. Classical InSAR reviews established the physical and algorithmic basis for measuring surface deformation from radar phase differences [2, 3]. Reviews of Earth-surface applications then consolidated InSAR as a deformation-measurement technology across geophysical and engineering settings [4, 5]. Persistent-scatterer and small-baseline time-series methods support monitoring applications in areas with repeatable radar phase histories [6, 7]. Later multi-temporal algorithms broadened this monitoring capacity across point-like and distributed scatterers [8, 9]. Public processing chains such as LiCSBAS have moved these products beyond specialist radar groups [10]. Regional public products now support studies of urban and deltaic subsidence [11, 12]. As these products enter screening workflows, the key question is how deformation layers should be translated into decision-facing exposure estimates without losing the support limitations recorded in the product. This translation step is both a signal-processing problem and a spatial-accounting problem. InSAR products do not observe all surfaces equally. Temporal decorrelation and surface change can reduce the validity of interferometric measurements [13]. In urban and peri-urban settings, vegetation, water, construction activity, geometric effects, coherence thresholds and product-specific quality masks can further structure the valid support of a public deformation layer [14]. Applied subsidence studies show that such deformation can be spatially uneven within cities and engineered landscapes [15, 16]. When a deformation map is intersected with population, built-up land or infrastructure layers, restricting the calculation to valid radar support defines a support-conditioned domain. That restriction is appropriate when the reported target is explicitly conditional on observed locations. A summary for a broader geographic domain must additionally report the population or built surface allocated outside that support. These are different accounting targets; missing support alone does not establish deformation at the omitted locations. Existing subsidence studies establish the importance of vertical land motion for cities, coasts and infrastructure. Coastal and urban studies link subsidence to exposure and flood risk [17, 18], while infrastructure and megacity analyses extend these questions to the built environment [19, 20]. Global and regional reviews document the geographic reach of the problem and the role of satellite interferometry [21–24]. Recent studies address observation support and population weighting at distinct scales. Xiong et al. quantify nationwide population, building and infrastructure exposure [25]. Yang et al. evaluate building-level observation adequacy using persistent-scatterer count, density, coverage and coherence [26]. Oelsmann et al. integrate vertical-land-motion observations for population-weighted coastal assessments, reporting population coverage and measurement and spatial uncertainties [27]. Building on these contributions, we ask how much a population-weighted support summary changes when the same population total is redistributed within

reporting units while the geographic domain, support field and population model remain fixed.	093 094
Published assessments also combine quality screening with geodetic comparisons.	095
The Mexico-wide assessment applied temporal-coherence screening and compared InSAR estimates with GPS observations [28], and Ohenhen et al. evaluated rates against 154 GNSS stations across 28 US cities [29]. Published Bangkok studies provide further leveling and CORS comparisons, with their spatial, temporal and station-specific qualifications detailed in Supplementary Section S3 [30–32]. Murray et al. also noted that motion between buildings may remain undetected and require additional observations [33]. These practices and stated limitations provide an essential basis for interpreting displacement measurements and their spatial support.	096 097 098 099 100 101 102 103 104
Population allocation introduces a further step when displacement products are combined with demographic data. Blackwell et al. estimated community exposure assuming uniform population within each community, while Ohenhen et al. assigned census-block population to subsidence classes using each block’s median InSAR-pixel rate [29, 34]. Both studies also used GNSS comparisons. Those comparisons assess displacement agreement; the allocation rules determine how measurements are translated into population summaries. We examine this latter step under a specified observation-support field, holding the domain, support and population total fixed. These published choices motivate the comparison without implying that either study mishandled missing observations.	105 106 107 108 109 110 111 112 113 114
This accounting question arises when independently useful datasets are integrated into an exposure summary. The radar product carries measurement support, a deformation screen carries a physical selection rule, and the exposure layer carries population or built-up surface. Their integration must distinguish allocation outside the analysis domain from allocation whose deformation status is unresolved. Reproducibility therefore requires recording when masks are applied and how they change the denominator, as well as which datasets are used.	115 116 117 118 119 120 121
The users of released InSAR products are often not the groups that generated the radar time series. A downstream analysis may combine an existing quality mask with a different population grid or reporting geography. Spatial support is part of that data model [35], and aggregation can alter the inferred quantity, as recognized by the modifiable areal unit problem [36]. Our matched-input comparison retains existing support information while testing whether within-unit population–support association is preserved or replaced by uniform allocation.	122 123 124 125 126 127 128
Areal interpolation and dasymetric mapping provide established approaches to allocating population within coarse units [37, 38]. A finer grid preserves more of the selected model’s spatial structure, without establishing household accuracy. Replacing the average population–support product with the product of separate averages omits within-unit covariance. This established identity explains a possible aggregation effect; it does not determine its empirical size. The questions are whether the effect persists away from boundaries, across population models and support definitions, and when missingness leaves some reporting units without an estimate.	129 130 131 132 133 134 135 136 137 138

139 We address this problem by comparing population-allocation rules under specified
 140 product-support fields. The proposed workflow keeps the geographic domain, radar
 141 observation support and allocated population or built surface separate until the final
 142 accounting step. We retain continuous mean area–pair support rather than using a
 143 majority threshold as the primary definition. The resulting support deficit asks how
 144 much of the selected model allocation is unsupported on average across the specified
 145 observations. It does not identify unique unobserved residents or confirm subsidence
 146 in locations without measurements. The earlier binary formulation is retained as a
 147 historical comparator in Section 2.3 and Supplementary Section S2.
 148 The audit addresses population-weighted accounting conditional on product sup-
 149 port. It does not assess whether displacement measurements are accurate where
 150 they exist. Figure 1 summarizes the comparison of uniform coarse allocation and
 151 native-model allocation under identical inputs, with threshold, allocation and
 152 partition sensitivity reported explicitly [39].



168 The contribution is an empirical evaluation of population-weighted support account-
 169 ing under matched inputs. Scale, origin and complete-interior comparisons isolate
 170 allocation effects from boundary handling. Crossed population-model and pair-
 171 selection tests, followed by a same-grid comparison of input support and final
 172 quality, establish how the effect depends on the chosen data and support defi-
 173 nition. Controlled missingness evaluates both error where an estimator remains
 174 computable and allocation left unestimated. Regional cases then test transfer. These
 175 comparisons identify when uniform allocation is adequate and when preserving the
 176 population–support association matters. The reproducibility package records the
 177 inputs, denominators and executable comparisons [40].

178 We report small effects, sign reversals and unestimated allocation alongside the pri-
 179 mary result. Physical interpretation requires geodetic and geotechnical evidence
 180 matched to the locations and quantities being assessed.

2 Materials and methods

2.1 Task formulation and notation

The workflow combines a declared geographic domain, reporting grid, support field and population or built-surface layer. It returns supported and unsupported allocation and their sensitivity to support and allocation choices. This accounting is downstream of displacement retrieval; the separate processing experiment below evaluates one specified LiCSAR/LiCSBAS chain.

We distinguish the geographic domain D (area of interest, AOI), support $s(x) \in [0, 1]$, and nonnegative exposure density $\rho(x)$. The primary Chao Phraya domain contains the 1,807,700 GHSL 3-arc-second pixels whose centers lie in the frozen finite footprint of 18,077 archived VLM cells. It is held fixed without velocity-threshold selection. Because the boundary inherits the finite source-product footprint, it does not represent complete coverage of the geographical delta. We retain this archived footprint as a fixed spatial reference for the allocation comparisons and for tracing changes from the earlier analysis.

The archived VLM GeoTIFF matches the Chao Phraya file in the global-delta dataset accompanying Ohenhen et al. [41, 42]. That study used WabInSAR. Our LiCSAR quality sample is a different processing chain and cannot be interpreted as the validity mask of the WabInSAR product. Moreover, the public GeoTIFF lacks an explicit stored-unit tag establishing the archived factor-of-ten conversion. We therefore retain the inherited strong-deformation selection only as a clearly labeled supplementary sensitivity. Neither that conversion nor a cross-product quality proxy determines the primary geographic population denominator.

For pair j , $M_j(x) = 1$ where unwrapped phase is finite, raster-valid and nonzero and, for the joint screens, byte coherence divided by 255 is at least q . The zero rule is product-specific [43]. We evaluate $q = 0.2, 0.3, 0.4$ and unwrapped validity alone. Integration uses each raster's actual GDAL transform and intersects masks before averaging; separately averaged masks would not yield their spatial intersection. For GHSL pixel p with area A_p , the reported support is

$$s_p = \frac{1}{NA_p} \sum_{j=1}^N \int_p M_j(x) dA. \quad (1)$$

Thus s_p is mean area-pair support at the exposure grid. It is distinct from the area whose individual locations pass a fixed number of pairs, from a binary majority classification at a cell center, and from final inversion validity. We assume each GHSL pixel's population or built-surface total is uniform within that pixel when integrating its intersections. Spherical rectangular areas use longitude and sine-latitude boundaries.

With nonnegative allocated exposure E_p , supported allocation is $S_E = \sum_p E_p s_p$, mean support deficit is $U_E = \sum_p E_p (1 - s_p)$, and total allocation is $T_E = \sum_p E_p$, so $S_E + U_E = T_E$. A pixel assigned 100 people and mean support 0.8 contributes 20 population-allocation units to U_E . These are averages over area and pairs, not

20 identifiable residents. Neither population-valued nor built-surface-valued deficits establish deformation, vulnerability or damage. The weighted support distribution is

$$F_E(t) = \frac{\sum_p E_p \mathbf{1}(s_p < t)}{T_E}, \quad \int_0^1 F_E(t) dt = \frac{U_E}{T_E}. \quad (2)$$

We report the full support distribution. Its integral is checked using sorted support values, not trapezoidal integration of the plotted 101-point curve.

2.2 Input schema and reproducibility contract

Each run requires stable cell identifiers, geometries and areas; a geographic domain declared before accounting; documented observation support; and population or built-surface quantities on a known spatial support. Any deformation selection additionally requires established units and a stated role. Raster masks, measurement polygons and point observations need distinct support definitions. A point is not repeatedly assigned the full population of its containing cell; an areal rule must first be specified and checked for overlap.

A clipped raster without its removed-support record cannot identify the omitted domain. Likewise, population without geographic support cannot be allocated, and infrastructure counts without asset metadata do not establish engineering risk. These input limits prevent a numerical total from being interpreted beyond its target.

Manifests record source versions, transforms, nodata conventions, established units, dates and checksums. Raster validity, coherence, final quality masks and polygon coverage remain separate fields. Intermediate outputs retain the domain, allocation rule, scale, origin and denominators, with per-pair, per-block or per-station records as applicable (Supplementary Sections S1, S4–S5 and S7). Transfers preserve extensive totals on the common domain.

2.3 Workflow architecture

The binary cell formulation separates deformation-screen membership from whether sampled support passes a majority cutoff (Supplementary Fig. S1). This partition defines a coarse-cell selection: a majority threshold does not identify permanently unobserved locations, and a selected cell does not locate residents over subsiding ground. We retain the original classification as a historical comparator. Continuous support on the fixed geographic domain defines the primary analysis.

Module 1 registers inputs to the common domain, retaining included area in partial boundary cells. Module 2 intersects validity and coherence masks before integrating support at native population resolution. Module 3 fixes geographic and population denominators independently of allocation; the historical deformation-selected partition remains in Supplementary Fig. S1 and Supplementary Section S2. Module 4 computes supported and unsupported allocation under uniform-area and native-model rules with identical totals. Module 5 tests sensitivity to support definition,

sampling, population model and reporting units. Recorded input selections define each run; thresholds are not chosen to maximize the deficit. The center, area-weighted and ancillary-weighted rules represent distinct choices within established areal-interpolation practice [37, 38]. Supplementary Section S6 states their information requirements and which comparisons are actually performed. Native GHSL allocation is the common model reference for these tests; comparisons with deformation imputation or multi-sensor retrieval would require a different target and additional observations. Extensive first and cross moments are conservatively transferred to a 50-m UTM integration grid (zone 47N) and repartitioned into 250-, 500-, 1,000- and 2,000-m cells. Each size uses four origins, with zero and half-cell shifts. Partial cells retain only included area; population is not extrapolated outside the domain. A 25-m recomputation checks integration sensitivity (Supplementary Sections S7 and S18). Within cell c , let $P_c = \int_c \rho \, dA$, $H_c = \int_c \rho s \, dA$, $A_c = \int_c dA$ and $Q_c = \int_c s \, dA$. Native-model reference deficit is $P_c - H_c$, while uniform allocation gives $P_c(1 - Q_c/A_c)$. Their difference is

$$U_{\text{uniform}} - U_{\text{native}} = \sum_c \left(H_c - \frac{P_c Q_c}{A_c} \right) = \sum_c A_c \text{Cov}_{A,c}(\rho, s). \quad (3)$$

Preserving H_c retains the population–support association. Center sampling instead uses the fine pixel at each coarse center, with zero support outside the domain. We report signed and absolute discrepancies relative to the native-model deficit and fixed AOI total. A complete-interior comparison excludes boundary-crossing cells, with separate subset denominators and containment checked against the transformed native-pixel boundary (Supplementary Section S7). The native reference is a population model, not household truth.

2.4 Study design and evidence roles

Chao Phraya supplies the primary allocation comparison, a separate Bangkok rectangle the connected processing experiment, and Fresno a published observation footprint. Los Angeles and Davis–Sacramento test additional product and regional transfer. These designs have separate support definitions and validation targets. In a fixed Bangkok rectangle (100.35–100.65° E, 13.65–14.05° N), 437 pairs from 2019–2020 with baselines at most 96 days connect 104 acquisition dates. LiCSBAS2 steps 11–16 use the aligned native inputs and fixed defaults without automatic threshold relaxation [10, 44]. Before pixel-quality inspection, 44 edges are randomly held out while preserving date connectivity; 393 training pairs undergo the same processing. Training time-series differences predict relative LOS displacement at the held-out edges, with training-linear and zero-displacement baselines on common evaluable observations. Referencing, exclusions and exact parameters are recorded in Supplementary S17.

Support strata and quality masks use training inputs only. Held-out dates remain represented in training, so shared dates and atmospheric phase restrict interpretation to internal edge reconstruction, not independent velocity accuracy or forecasting. Separately, legacy 28-pair and matched-stack support are compared with full-stack quality, with GHSL counts transferred by source-area fractions. Coherence contributes to mask construction, and the legacy period differs. This evaluates the specified LiCSAR/LiCSBAS chain, not the WabInSAR raster; matched independent Bangkok GNSS validation remains unavailable.

A fixed Fresno rectangle (119.9–119.7° W, 36.6–36.85° N) tests transfer to the DWR 2026Q1 observation-cell footprint [45, 46]. We clip and union all 38,738 intersecting measurement polygons, then allocate GHSL population once to their nonduplicated intersections. Scale/origin comparisons use UTM zone 11N. The polygon layer supplies coverage, not per-cell velocity or uncertainty; its 2015–2026 observation record also differs from the 2020 population epoch. Artificial masking is restricted to the published footprint.

Population agreement and external transfer designs.

Fresno’s irregular 2020 Census blocks use GHSL allocation normalized to P1 totals and known DWR rates for 1 October 2020–1 October 2021. Random masks and one or four contiguous holes remove 10, 30 or 50% of known-rate cells, with 30 repetitions at -3 , -5 and -10 mm yr $^{-1}$. Native/area prevalence and unit-median rules share masks; undefined allocation, common-computable errors and whole-domain bounds are reported separately (S15).

Five-band spatial holdouts with 0-, 1- and 3-km buffers test training-only total recalibration against ACS, not original population-model training. Shared-date omissions assess dependence in the 224-pair pool without velocity re-inversion (S20). Eighteen training-only quality rules evaluate coverage and residuals on 44 excluded edges; held-out errors do not select thresholds (S21).

Fixed Los Angeles and Davis–Sacramento rectangles test three cutoffs, four scales and four origins using archived ARIA products [47, 48] (S29). Fresno municipal land use supplies strict/broader residential eligibility with unmapped areas retained [49] (S27). Archived NGL GNSS LOS comparisons retain both reference frames and three sampling radii, excluding provider and location-selected local controls from all validation metrics. Temporal-model and source-version limits are retained; this external-product check neither refits daily GNSS nor validates Bangkok (S30).

2.5 Exposure layers and allocation

GHSL supplies consistent public population and built-surface layers across the study domains. Comparing uniform-area and native allocation isolates a spatial assumption under identical inputs; neither represents household-level validation. The resulting quantities are model allocations rather than official demographic or loss estimates.

We compare GHSL R2023A epoch-2020 population at 3 arcsec with WorldPop’s Thailand 2020 unconstrained UN-adjusted population-count product at 3 arcsec [50, 51]. Source counts are transferred by spherical source-area fractions, not by

interpolation of density values as if they were counts. Native totals, jointly valid coverage and common-total normalized results are all reported. GHSL 2015/2020 epoch comparisons hold resolution at 30 arcsec. Neither population product is local census truth, and epoch differences combine demographic and model effects.

Crossed aggregation sensitivity.

We cross GHSL and WorldPop 2020 with legacy 28-pair and expanded 224-pair support at $q = 0.3$. All combinations use the same two-model common-valid domain and GHSL total, with 1-km cells at four origins and conserved first and cross moments. Differences are reported relative to both the native deficit and common population total. The 25-m check, all 16 cases and pair-selection details remain in Supplementary Section S18; changing pair number also changes composition. WorldCover 2021 v200 supplies 10-m land classes [52]. Their fractional areas within GHSL pixels inherit the within-pixel uniform-allocation assumption. Tree-class allocation does not identify residences, and land-cover, population and radar epochs differ.

2.6 Robustness and bounded-evidence protocol

The legacy 28-pair sample from 2016–2022 is retained as a traceable reference to the archived analysis, with $q = 0.3$, the middle of the three evaluated coherence cut-offs, used for the main display. These reference settings do not establish an optimal threshold or final-product quality; the expanded sample and matched final-mask comparison assess dependence on the input sample and support definition. The expanded pool selects eight pairs per year–quarter stratum (224 total) using dates and baseline classes before outcome extraction. Nested ranks yield 28, 56, 112 and 224 pairs; 100 stratified draws without replacement per size assess conditional sampling variation. A pre-extraction rank amendment and both rank plans are archived. Year, quarter, baseline and shared-date omissions test composition and dependence. These comparisons do not estimate independent-observation confidence bounds or uncertainty outside the fixed pool (Supplementary Sections S16 and S20). Controlled masking uses a separate fixed Bangkok rectangle (100.35–100.75° E, 13.55–13.95° N). Random, spatially smoothed and positive/negative log-density-associated scores remove 10%, 30% or 50% of fine pixels, with a fixed seed and 100 repetitions. Uniform-area, center and built-area-weighted rules share masks and coarse totals; built weights fall back to area where zero. Grids contain 5, 10 or 20 native pixels at two origins, not exact metric cells. Errors use the known removed allocation U_{ref} . Percentile ranges describe introduced-mask variation conditional on the fixed inputs, not population or InSAR accuracy. Built weighting shares information with GHSL and is not independent validation. We also construct three synthetic hazard fields: positive density association, negative association and an independent spatial field. Rates of -10 or 0 mm yr^{-1} define known threshold membership. For observed threshold allocation T_{obs} and allocation U in pixels missing under a binary final mask, the identification interval is $[T_{\text{obs}}, T_{\text{obs}} + U]$. Its endpoints assign all missing allocation outside or inside the class. Here U is binary-mask missing allocation, not the continuous mean area–pair deficit.

The bound is conditional on the observed rates and population model; it is neither a physical prediction nor a confidence interval.

3 Results

3.1 Lead-case support accounting

Figure 2 compares native-model and uniform allocation under the same support field, geographic domain and allocated totals.

The fixed Chao Phraya footprint contains 22.501 million GHSL population units and 1253.6 km² of built surface. At $q = 0.3$, native allocation gives $U_E = 3.020$ million population units (13.42%) and 325.6 km² of built surface. Population is concentrated in relatively well-supported portions of the footprint (Supplementary Fig. S14). Consequently, area-, built-surface- and population-weighted support distributions differ (Fig. 3).

Table 1 Native-model support deficits.

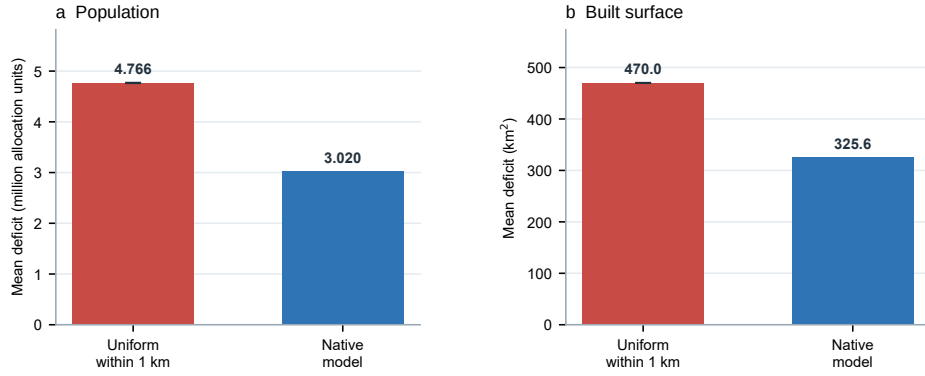
Support rule	Population (million units)	Population (%)	Built surface (km ²)	Area (%)
Valid unwrapped	0.268	1.19	31.9	17.89
Joint, $q = 0.2$	1.665	7.40	197.5	57.44
Joint, $q = 0.3$	3.020	13.42	325.6	68.77
Joint, $q = 0.4$	4.891	21.74	464.6	76.83

Unwrapped validity alone and the coherence-screened quantities are substantially different (Table 1). Raising q from 0.2 to 0.4 changes the mean population deficit from 1.665 to 4.891 million. This range is a definition sensitivity, not a confidence interval. The weighted support curves retain information lost by applying one majority threshold to every coarse cell.

Supplementary Section S2 retains the original 3.77-million statistic and input lineage. Its center-majority selection and domain differ from the present mean deficit; the new value is not an estimate of the same quantity or a count of confirmed subsidence exposure.

3.2 Sensitivity to support definition, allocation and population models

With support, native population and AOI total held fixed, uniform allocation within 1-km cells yields 4.763–4.770 million mean population units in deficit, compared with the native-model reference of 3.020 million. This is 57.7–57.9% higher across the four origins. Aggregate allocation discrepancy grows across the examined 250–2,000-m scales (Supplementary Fig. S15). The center approximation also changes with scale and origin; its variability cannot be interpreted as a measurement uncertainty distribution.



The signed discrepancy follows the population–support covariance in Eq. 3. Positive aggregate covariance means uniform redistribution moves population away from better-supported locations, although individual cells can have either sign. Using 25-m instead of 50-m integration changes the 1-km estimate by at most 0.00024 percentage points of AOI population. A separate pixel-center diagnostic verifies the identity and its Cauchy–Schwarz bound (Supplementary Sections S8–S9).

The 1-km effect persists when all boundary-crossing cells are excluded. Across four origins, the complete interior cells contain 99.27–99.49% of domain population, and uniform allocation exceeds their own native-model deficit by 58.24–58.56%. Interior cells account for 98.55–98.87% of the full-domain signed uniform-allocation difference. Thus boundary treatment does not explain the primary allocation effect. Supplementary Section S7 separates complete, partial and outside-center cells and reports both methods under each origin.

On pixels jointly valid in the four population/epoch layers, the $q = 0.3$ deficit is 13.37% for native GHSL 2020 and 23.20% for WorldPop 2020. Joint coverage excludes 142.9 thousand GHSL population units from the full primary domain. Equalizing the two models to the same joint-domain total gives 2.989 versus 5.186 million mean deficit units, so total-population differences do not explain the discrepancy. At fixed 30-arc-second resolution, GHSL’s 2015 and 2020 shares are 20.86% and 20.28%. These epoch sensitivities are smaller than the native-model contrast here, without establishing which model is accurate (Supplementary Fig. S16).

Crossing population model and pair selection.

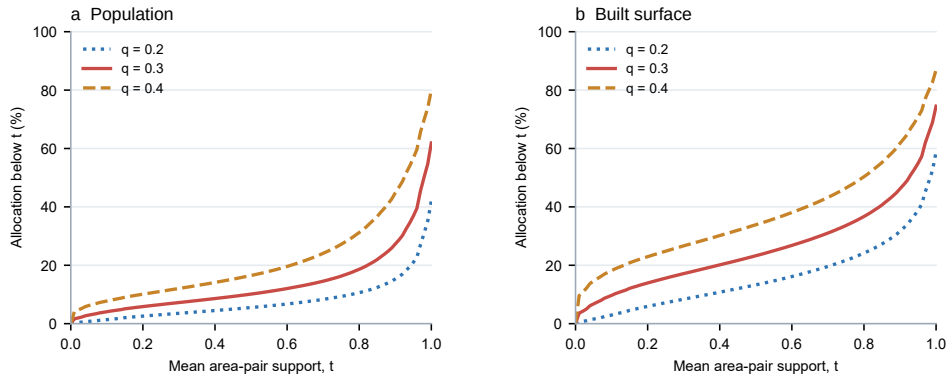
On the fixed GHSL–WorldPop common domain, both models are normalized to 22.358 million population units. At 1 km, uniform allocation exceeds the native-model deficit by 55.38–55.64% for GHSL with the legacy 28 pairs and 51.29–51.51% with the expanded 224 pairs. The corresponding WorldPop differences are 9.03–9.07% and 8.64–8.68%. All 16 model–sample–origin combinations have positive differences; normalized by the common total, these span 7.40–7.53 percentage points for GHSL and 2.09–2.13 for WorldPop. The direction therefore persists across these inputs, while its magnitude is strongly population-model dependent. The original

approximately 58% difference is specific to its full-domain GHSL case. Changing 28 to 224 pairs changes selection and composition as well as sample size; the comparison is not an isolated sample-size effect. Supplementary Section S18 reports every origin and the numerical convergence check.

The Fresno population comparisons separate source totals from spatial allocation. Against 2020 Census blocks, total-normalized 1-km GHSL and WorldPop allocations have weighted absolute percentage error (WAPE; summed absolute unit discrepancies divided by the Census total) of 57–59%, below an area-only baseline (Supplementary Section S24). A 100-m comparison normalizes GHSL and the California population surface (CA-POP) to the same Census block counts; the outside-footprint allocation differs by 0.143 pp, with mixed tract-level directions (Supplementary Section S25). CA-POP uses the same Census counts as inputs, so this is an allocation sensitivity.

Against ACS 2017–2021 estimates for 319 complete Fresno block groups, Census P1, GHSL and WorldPop totals differ by -0.36% , $+10.57\%$ and $+6.03\%$, respectively. Census falls within the aggregate ACS 90% margin of error, without validating household locations. Most block groups are smaller than the 1-km source pixels, so source-cell uniformity remains material. Buffered spatial recalibration gives similar in-sample and held-out errors (Supplementary Sections S20, S26); these reporting-unit checks do not establish fine-scale accuracy.

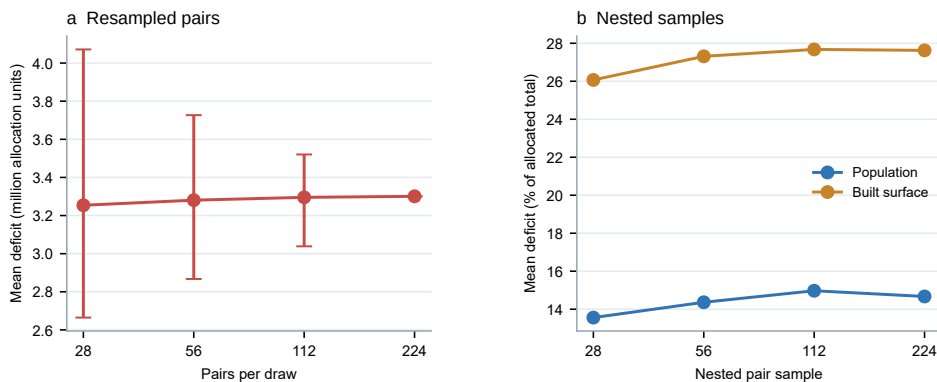
Municipal residential eligibility leaves about one-third of model allocation unclassified. Its restricted common domain covers 89.13% of complete Census population, below the exploratory 90% reporting gate. Reduced deficits on that domain therefore support only a limited-domain sensitivity (Supplementary Section S27).



3.3 Sampling uncertainty and its scope

Figure 4 distinguishes conditional variation over draws from the fixed nested selections.

The legacy 28-pair date graph contains 54 dates, 26 components and no independent cycles. It cannot stand in for a complete time-series inversion network. The 224-pair sampling graph is also not fully connected: it has 225 dates and 29 components. In contrast, the 2016–2022 catalogue graph has 288 dates and one component. Catalogue-level connectivity does not establish pixel-level validity. The nested 28-, 56-, 112- and 224-pair means give population deficits of 3.051, 3.232, 3.369 and 3.301 million, respectively (Supplementary Fig. S4). The nested 28-pair subset is selected from the expanded pool and differs from the original 28 pairs used in the primary 3.020-million calculation. Legacy leave-one-year-out results range from 2.820 to 3.181 million. Within the fixed 224-pair pool, the central 95% range of the 100 stratified 28-pair draws is 2.665–4.072 million. That conditional sampling range is wider than the differences between several individual nested means and cannot be converted into a confidence interval for final product quality. Longer-baseline groups have larger mean deficits in both samples. Differences between sample sizes also reflect changes in baseline and seasonal composition. Deleting each shared acquisition date changes the $q = 0.3$ equal-quarter population deficit by -0.386 to $+0.165$ pp. These are dependent support sensitivities, with no velocity re-inversion or confidence interval (Supplementary Section S20).



The checks are summarized by the alternative explanation they address (Table 2). Their different targets preclude treating them as a single validation score.

3.4 Descriptive ranking stability

Spatial rankings are more stable than the totals: across 539 eligible fixed geographic blocks, nested 28-, 56- and 112-pair deficit-amount ranks correlate 0.9940, 0.9992 and 0.9994 with the 224-pair comparator. Top-decile overlaps nevertheless change (Supplementary Fig. S5). This is descriptive stability, not validation of a measurement-priority rule.

Top-decile Jaccard overlap for deficit amount is 0.8000, 0.8621 and 0.9286 for nested 28, 56 and 112 pairs against 224 pairs. Supplementary Fig. S6 reports amount and

Table 2 Checks on the accounting result and their interpretation.

Check	Result or comparison	Interpretation
Support definition	Same-grid uniform-minus-native shift: input support 3.066–3.189 pp; final mask 0.108–0.117 pp	Input support is not final-product validity
Scale and grid origin	Complete-interior 1-km difference: 58.24–58.56% of native deficit	Boundary handling does not explain the primary effect
Population and pair selection	Positive shift in all 16 GHSL/WorldPop and 28/224-pair origin cases	Direction persists; magnitude depends on the population model
Controlled missingness	At 30% removal, irregular- block prevalence leaves 0.20% unestimated for random pixels versus 23.01–28.64% for holes	Conditional error and estimator availability must be reported together
Sampling dependence	Fixed-pool repetitions and shared-date omissions	Conditional sensitivity; not independent-observation confidence bounds
Processing and transfer	Excluded pair edges, DWR geometry and external California GNSS	Different evidence targets, summarized in Supplementary Table S29

share rankings. High all-block correlation can coexist with changed short-list membership; monitoring priority would additionally require deformation, consequences and information value.

3.5 Failure-mode experiments and estimator availability

For random and spatial masks unrelated to density, aggregate mean bias is small across repetitions, while the center method has larger sample-to-sample variation. Density-associated masking changes the sign and size of the error (Supplementary Fig. S17). At 30% removal and 10-pixel cells with zero origin, uniform allocation underestimates the known removed population by 14.17% when high-density locations are preferentially removed and overestimates it by 120.35% when low-density locations are preferentially removed. Large relative errors occur when the removed population is small. We also report errors normalized by the total test-domain population.

Built-area weights reduce error in these GHSL-based scenarios but retain nonzero errors and shared-input dependence. The synthetic hazard experiments additionally show that support deficit alone does not identify threshold-exposed population: the unknown allocation can contain very different hazard shares under the three constructed velocity fields. The conservative identification interval contains the known truth by construction; its width expresses missing information rather than recovered

motion. Fresno masking outputs repeat the same protocol and are supplied in the
 Supplementary Information.

Small conditional errors can coexist with substantial unestimated allocation. At
 30% removal, 10-pixel units and -5 mm yr^{-1} , prevalence leaves less than 0.0001%
 unestimated under random masks but 11.35–28.66% under structured masks. On
 the common-computable domain, prevalence has smaller local L1 error than median
 classification for three mechanisms; the median rule is better under low-spread
 removal. Whole-domain bound widths remain 24.12–34.90 pp (Supplementary
 Sections S12, S14).

Actual irregular Census blocks show the same availability problem. At 30% removal,
 prevalence leaves 0.20% unestimated under random masking, versus 23.01% under
 one contiguous hole and 28.64% under four holes. The median rule leaves 9.41%,
 26.80% and 32.61%, respectively. Its whole-block classification also differs from the
 fine-allocation target before masking. Complete error and bound results are given in
 Supplementary Section S15.

For the actual LiCSBAS final mask, unsupported allocation is 0.2944% of the fixed
 domain total. At LOS rate $\leq -5 \text{ mm yr}^{-1}$, observed class allocation is 0.2797% and
 its whole-domain share is bounded by [0.2797, 0.5741]%, conditional on the observed
 rates and population model. The 0.2944-pp interval is narrow relative to the domain
 total but its upper endpoint exceeds twice the observed class share. Both end-
 points are attainable by changing only the unknown class assignments. Thus a small
 unsupported share can still matter for a rare threshold class; no motion is recovered
 in the gaps (Supplementary Section S13).

3.6 Regional and product transfer

Figure 5 compares the allocation effect and population-weighted coverage under the
 declared regional support definitions.

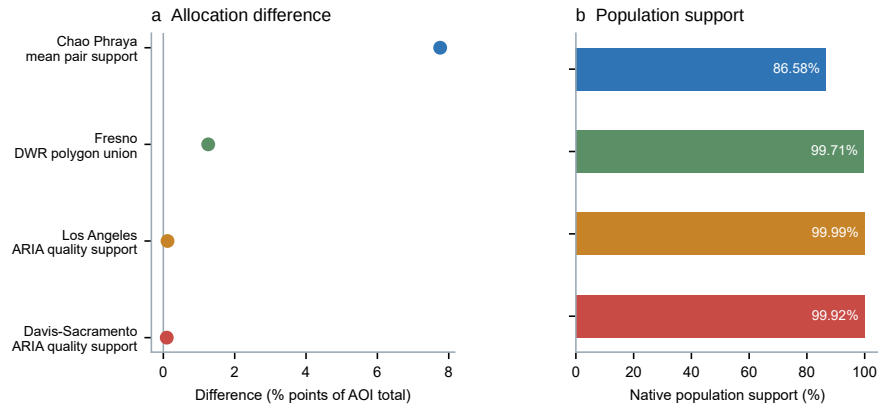
In Fresno, the DWR observation-cell union covers 78.49% of the 495.51-km² rect-
 angle but contains 99.71% of its GHSL allocation. Of 598098 allocation units, 1710
 fall outside the footprint. At 1 km, uniform allocation gives 9022–9556 outside-
 support units, an additional 1.2–1.3 pp of the AOI total. Its large relative multiplier
 therefore has a small baseline. Center sampling also varies with grid origin (Supple-
 mentary Fig. S19).

On the complete Census-block domain, the Census-population-weighted DWR
 area-coverage share is 99.19%, versus 99.74% for native GHSL allocation (Sup-
 plementary Section S23). This normalized-share contrast reflects relative spatial
 weights, whereas the 12.20% GHSL–Census total difference is a separate diagnostic.
 Multiplying all weights by one constant leaves either share unchanged.

A separate DWR rate example uses Census-block median observed vertical-rate
 classes. The annualized six-window 2015–2021 median covers blocks containing
 95.70% of complete-block population, of which 64.99% falls below -5 mm yr^{-1} .
 Different time-window choices change classification; this transfer does not validate
 Bangkok LOS rates (Supplementary Section S28).

Across 96 fixed-region scale/origin/cutoff designs, uniform-minus-native differences
 are only 0.026–0.186 pp in Los Angeles and 0.051–0.372 pp in Davis–Sacramento.

691 The positive direction transfers, but these small effects do not generalize the
 692 approximately 58% primary relative difference (Supplementary Section S29).



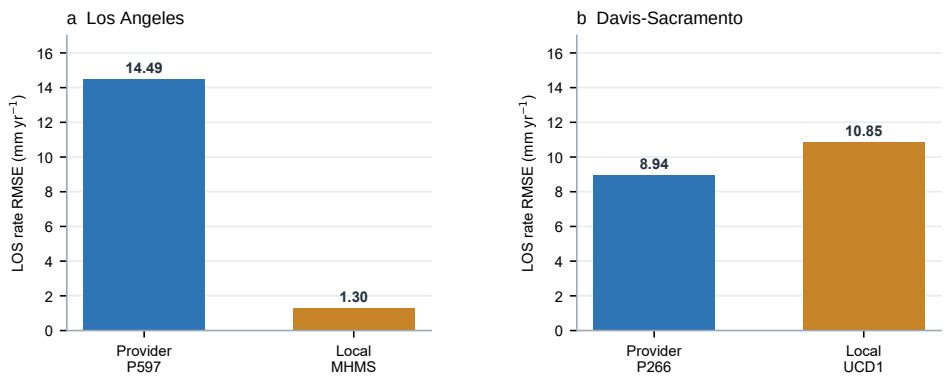
3.7 Processing checks and external evidence limits

693 Full-stack quality rejection leaves 431 of 437 pairs for inversion; the training branch
 694 retains 382 of 393 inputs. The full quality mask accepts 97.02% of pixels and 99.71%
 695 of the small area's allocated population. Among pixels with matched-input mean
 696 support below 0.2, 83.34% still pass the fixed multi-index mask; the 0.8–1 group has
 697 99.72% acceptance. Low coherence-screened support is therefore not equivalent to
 698 final processing rejection (Supplementary Fig. S18). Neither acceptance nor rejection
 699 is independent physical truth.

700 Matched-input support bins also differ in median velocity spread (1.343 versus 1.185
 701 mm yr⁻¹) and inversion residual (0.673 versus 0.368 mm, lowest versus highest
 702 bins). These are quality associations, not a binary validity surrogate; legacy-sample
 703 confusion counts and period differences are reported in Supplementary Section S17.
 704 All 44 held-out edges met the prediction and reference requirements. Pooled held-
 705 out LOS displacement root-mean-square error (RMSE) is 0.984 mm across evaluable
 706 observations. Training-mask accepted and rejected locations give 0.699 and 5.150
 707 mm, respectively; the lowest and highest training-support bins give 2.338 and 0.412
 708 mm. On common evaluable observations, time-series differences, training linear
 709 velocity and a zero-displacement baseline give 0.984, 11.751 and 11.784 mm. The
 710 held-out pairs connect dates already represented in training. Shared acquisition
 711 dates and atmospheric phase limit these residuals to internal reconstruction; they
 712 do not measure independent velocity accuracy. Exclusions and per-pair results are
 713 retained in Supplementary Section S17.

714 On the same processing grid and GHSL field, uniform allocation within 1-km cells
 715 increases the deficit by 3.066–3.189 percentage points of AOI population for mean

437-pair input support, versus 0.108–0.117 points for the final quality mask. Native deficits are 3.4539% and 0.2944%, respectively. Direct source-cell intersections verify the geometry (Supplementary Section S19). This contrast reflects different support definitions, not an error comparison or observed motion in gaps. Training-only coherence filtering from 0.2 to 0.4 reduces accepted allocation from 99.40% to 95.34% and pooled excluded-edge RMSE from 0.491 to 0.388 mm. At 0.7 only 41.04% remains. Other metrics have nonmonotonic responses; all 18 fixed-inversion post-filters are reported in Supplementary Section S21, without interpreting residuals as absolute velocity accuracy. External GNSS agreement is mixed. Native-pixel RMSE in Los Angeles is 14.49 mm yr^{-1} under provider reference P597 and 1.30 under location-selected local control MHMS (11 validation stations). Davis–Sacramento gives 8.94 under P266 and 10.85 under UCD1 (four stations). Controls are excluded from validation; every sampling radius and both frames are retained. Archived source-version conflicts, temporal-model differences and station dependence limit interpretation. These results cannot be transferred to our Bangkok inversion (Supplementary Section S30). Figure 6 retains both reference-frame choices in each region.



Supplementary Fig. S20 and Supplementary Table S29 distinguish accounting checks, internal reconstruction and external-product comparisons from the unresolved physical validation of the Bangkok inversion.

783 4 Discussion

784

785 4.1 Earth-science informatics implication

786

787 The central implication is that public InSAR observability should be treated as part
788 of the information product when deformation maps are translated into exposure
789 estimates. A support mask specifies the observation domain and support available
790 to a downstream analysis. The proposed audit links that support to a traceable
791 population or built-surface allocation quantity.

792 The matched-input comparisons establish a material change in population-weighted
793 support summaries when fine spatial allocation is replaced by uniform coarse allo-
794 cation. The effect persists in complete interior cells, so boundary handling does not
795 explain the primary result. Crossing population models and pair selections gives a
796 consistent positive direction across the 16 tested cases, but the relative difference is
797 much smaller with WorldPop than with GHSL. The approximately 58% difference
798 is specific to the primary GHSL case; the crossed comparisons show how the mag-
799 nitude changes with the population distribution. It does not establish a universal
800 correction factor.

801 Measurement precision and coherent-scatterer reliability do not fully describe uncer-
802 tainty in exposure accounting. A product can be accurate where it observes while a
803 population summary remains conditional on that support. Reporting the continuous
804 support deficit makes explicit the mean population allocation unsupported across
805 the specified interferogram pairs. This complements the quality information supplied
806 by the radar processor.

807 For city analysts, infrastructure planners and hazard-screening researchers, the
808 support mask and population-allocation rule must remain explicit when a public
809 deformation layer is combined with demographic data. Supported and unsupported
810 totals provide a shared interpretation for radar specialists and downstream users
811 who do not revisit the full processing chain.

812 Reusable products therefore need machine-readable support semantics as well as
813 values. Velocity, coherence, validity and population belong to different data layers.
814 Keeping them separate distinguishes zero exposure from unsupported allocation and
815 allocation outside the domain; the final table should preserve those meanings.

816 Related support-and-allocation questions arise when other Earth-observation prod-
817 ucts are combined with human-exposure layers or administrative boundaries,
818 including flood, fire and coastal-impact screening. Such applications require their
819 own physical targets and tests; the present experiments establish results only for the
820 stated InSAR and population inputs.

821

822 4.2 Relationship to radar and signal-processing practice

823

824 From a signal-processing perspective, the support deficit is a downstream con-
825 sequence of non-uniform measurement support. InSAR observability depends on
826 the ability of the processing chain to retain coherent or otherwise valid measure-
827 ments through time. When that support is spatially structured, the product is not
828 a uniform sample of the chosen geographic domain. The audit therefore extends

a familiar signal-processing concern, measurement support and missingness, into population-weighted support accounting. This extension matters because missing support is not spatially neutral. Water bodies, vegetation, rapid surface change, construction disturbance and low-coherence surfaces are geographically structured. Urban deltas and coastal plains can combine dense built-up areas, waterways, soft soils and construction zones. Semarang illustrates a coastal-urban setting in which subsidence and exposure intersect [53]. Jakarta provides a related example where time-series radar has been used to measure coastal urban subsidence [54]. Wuhan and Karapinar show that subsidence settings can also vary across inland urban and basin contexts [55, 56]. If weak observability is spatially correlated with population allocation, a support mask can change the population-weighted support summary. The covariance relationship explains the primary overestimation and the opposite errors under density-associated masking. Small aggregate biases under density-independent masks identify conditions in which uniform allocation can be adequate. Preserving finer model structure is useful only insofar as that structure is appropriate to retain; agreement with a native-model reference does not establish household accuracy. Measurement support and measurement value remain distinct. Low support does not itself indicate subsidence. The primary geographic domain is therefore fixed without a velocity threshold, while the support field records mean area-pair availability. A physical deformation screen would require independently interpretable rate measurements. Reconstructing deformation in missing locations would require additional observations or assumptions about geology, groundwater, land cover, loading and spatial correlation. Those assumptions may be justified locally but cannot be inferred from a support deficit. Here, numerical correctness is assessed conditionally on specified inputs without asserting deformation in gaps. Ascending/descending geometries, local GNSS, leveling, piezometry or building observations could support a separate validation or attribution analysis. A verified deformation field would then permit population accounting within physical threshold classes, with unresolved locations still reported. Choosing new measurement sites would additionally require information value and local consequences.

4.3 Use for infrastructure and power-system screening

Although the lead claim concerns population and built-surface allocation, the workflow is extensible to infrastructure layers. For electrical-engineering or infrastructure-resilience applications, E_p could be replaced by substations, transmission-line length, distribution corridors or other critical assets. Such use would require asset-specific interpretation because infrastructure risk depends on design, age, redundancy, foundation conditions and maintenance. With an asset-specific support definition, the audit could report assets associated with weaker observation support. The present experiments do not evaluate those assets, failure probability or a power-system risk model.

875 An infrastructure application would first account for support at asset locations or
876 corridors, then combine it with engineering observations, asset criticality and the
877 value of additional information. The audit can identify incomplete radar support;
878 vulnerability and intervention priorities still require engineering assessment.

879

880 **4.4 When the audit should and should not be used**

881

882 The audit is most appropriate when a public deformation product is being reused
883 for screening and the downstream question involves exposure translation. It is useful
884 when the deformation product includes valid-support information, when exposure
885 layers can be allocated to the analysis support, and when the analyst needs to know
886 whether a visible-only summary has changed the denominator. Before additional
887 local validation, it can describe where the selected population model is associ-
888 ated with weaker support. It cannot, by itself, determine where measurement or
889 intervention would deliver the greatest benefit.

890 The audit is less appropriate when the question is already a dense site-level diagno-
891 sis. If a study has local leveling, GNSS, borehole, hydrogeological and engineering
892 observations over the same domain, then the main uncertainty may no longer be
893 public-product observability. The workflow can still document product support, but
894 it should not replace the stronger local evidence. The audit is also inappropriate
895 when the source product has already discarded invalid areas without preserving the
896 mask, because the omitted observation support can no longer be reconstructed reli-
897 ably. In that case, the honest output is a data-limitation statement rather than an
898 unsupported-allocation estimate over that missing domain.

899 Method choice follows the available information. A complete-case overlay describes
900 allocation inside its observed domain. A center rule is inexpensive but can miss sub-
901 cell structure and boundary coverage. Uniform-area transfer is appropriate when
902 its uniformity approximation produces acceptably small error for the intended sum-
903 mary. Building weights add ancillary assumptions and can share inputs with the
904 population model. Imputation, multiple sensors and ground measurements add
905 information about deformation and require their own evaluation. Our experiments
906 compare population-allocation rules under specified support fields.

907

908 **4.5 Limitations**

909

910 The strongest physical limitation is the absence of matched independent GNSS
911 or leveling validation for the Bangkok inversion. A local feasibility audit found
912 that CUSV and CUUT occupy the same InSAR pixel and lack an independently
913 established datum shared with the inversion (Supplementary Section S17). Con-
914 trolled masks test accounting against conditional references; excluded pair edges test
915 internal reconstruction. California GNSS supplies independent observations for a dif-
916 ferent product and different regions, with mixed results under two reference-frame
917 choices. None identifies deformation in permanently unsupported Bangkok locations
(Supplementary Fig. S20; Supplementary Table S29).

918 This limitation restricts physical interpretation. A support deficit cannot be con-
919 verted into confirmed affected population without evidence about deformation at
920

the relevant locations. The controlled experiments instead establish what changes when the same inputs are combined under different allocation rules. The primary Chao Phraya domain inherits the finite source-VLM footprint, so its conclusions do not establish coverage of the whole delta or extend to locations outside that footprint. The additional rectangles likewise do not establish a worldwide magnitude of the effect.

The original majority-observable partition simplifies a continuous support problem. The revised primary analysis therefore retains support fractions and the full population-weighted support distribution. This change addresses threshold information loss, but it does not make coherence-screened pair support equivalent to reliable velocity or to a final processing mask. Sampling variation remains conditional on the archived pair pool; shared acquisitions and atmospheric signals also limit the interpretation of held-out edge residuals. Pair number, seasonal composition and temporal baseline are not isolated experimental factors.

A further limitation concerns population and built-surface data. GHSL products are appropriate for a reproducible public-data workflow, but they are not substitutes for local census, cadastral or infrastructure inventories. Population grids can misallocate people in rapidly changing urban areas, and built-up products can differ in how they represent informal settlement, industrial surfaces or mixed land cover. WorldPop, Census, CA-POP and ACS comparisons quantify particular disagreements at stated reporting supports, but they do not fully propagate population-product uncertainty or establish household locations. CA-POP shares Census inputs, and land-cover or built-area weighting can share information with GHSL. The municipal residential classification is incomplete; results on its restricted common domain are not population-wide validation.

Finally, the workflow is designed for screening rather than causal attribution. Subsidence can reflect groundwater withdrawal, sediment compaction, loading, hydrocarbon extraction, tectonic motion or other processes. National-scale assessment shows that city-level subsidence patterns can differ across hydrogeological and urban settings [57]. Policy-oriented coastal work further shows that the benefits and priorities of subsidence control depend on local exposure and management context [58]. Mechanistic attribution would require hydrogeological, geological and geotechnical evidence beyond the current data stack. Further physical interpretation requires independent deformation observations matched to the lead domain and stronger local population or asset data.

Earlier estimates and source attribution.

The submitted 3.772492-million center-majority total is reproducible, but it is not a verified count of people exposed to subsidence in missing areas. The earlier center-majority statistic differs from the mean support deficit defined here; their relationship and input provenance are documented in Supplementary Section S2. We have removed the exploratory Iran branch from quantitative applications. Its archived rate and mask correspond to the 2014–2020 product associated with Haghshenas Haghighi and Motagh (2024). No Iran result or new/old Iranian product comparison contributes to the quantitative conclusions. Supplementary Section

967 S3 documents the attribution correction and distinguishes the newer products of
968 Payne et al. [12, 59, 60].

969

970 5 Conclusions

971

972 Spatial aggregation changes population-weighted InSAR support summaries
973 by omitting the within-cell population–support cross moment. Fixed-input and
974 complete-interior comparisons establish this effect. Its direction persists across
975 the tested population models and pair selections, while its magnitude depends
976 on the inputs and study area. Controlled missingness identifies conditions under
977 which the error reverses, aggregate bias is small or substantial allocation remains
978 unestimated. The published-support and regional comparisons show that transfer
979 also depends on the support definition; external GNSS agreement differs between
980 regions. Population-weighted reporting should therefore specify its domain and
981 support field, preserve extensive quantities and their cross moments, and report sen-
982 sitivity to allocation and sampling. Deformation in unsupported locations remains
983 unknown.

984 The contribution is a reproducible support-accounting workflow. Its use in screening
985 requires a declared domain, population model, support field and allocation rule. A
986 summary restricted to observed support should be labelled accordingly. Site-level
987 diagnosis and intervention decisions still require denser geodetic, demographic and
988 geotechnical evidence.

989

990 Data Availability

991

992 Public input data are obtained from COMET LiCSAR, GHSL, WorldPop, ESA
993 WorldCover, the U.S. Census Bureau, California DWR, the archived ARIA Cal-
994 ifornia products, NGL and City of Fresno municipal GIS. Exact source-product
995 identifiers, acquisition methods and input checksums are documented in the Supple-
996 mentary Information and accompanying reproducibility materials. Source-provider
997 terms govern access to and reuse of third-party inputs. Frozen processed analysis
998 inputs, full reference and replicate tables, figure source data and the manuscript
999 number ledger are publicly available with the versioned scientific reproduction
1000 package.

1001

1002 Code Availability

1003

1004 The scientific reproduction package is publicly available at the fixed GitHub release
1005 [v0.6.3](#). It contains the experiment code, configurations, input manifests and check-
1006 sums, processed analysis inputs, full reference outputs, environment records and
1007 reviewer reproduction instructions. Companion assets supply portable inputs and
1008 selected complete source products for S15, S20–S21, S27, S29–S30, with file-level
1009 checksums. The complete ACS files and official Census source archives used in S25
1010 remain available at v0.4.1. The reviewer guide provides reproduction and numerical-
1011 comparison commands for the processed inputs and reference outputs retained from
1012

release v0.6.0. The current manuscripts, two portable local audits and a section-to-output index are identified separately under `revision_2026/submission_20261003_final/`; these current documents supersede earlier manuscript snapshots in the archive. Other provider-controlled original products and the complete interferogram processing stacks are obtained separately through the documented acquisition routes. The earlier DOI [10.5281/zenodo.21444768](https://doi.org/10.5281/zenodo.21444768) archives v0.3.0; a revised DOI deposit remains pending.

References

- [1] Torres, R. *et al.* Gmes Sentinel-1 mission. *Remote Sensing of Environment* **120**, 9–24 (2012).
- [2] Bamler, R. & Hartl, P. Synthetic aperture radar interferometry. *Inverse Problems* **14**, R1–R54 (1998).
- [3] Rosen, P. A. *et al.* Synthetic aperture radar interferometry. *Proceedings of the IEEE* **88**, 333–382 (2000).
- [4] Massonnet, D. & Feigl, K. L. Radar interferometry and its application to changes in the earth’s surface. *Reviews of Geophysics* **36**, 441–500 (1998).
- [5] Buergermann, R., Rosen, P. A. & Fielding, E. J. Synthetic aperture radar interferometry to measure earth’s surface deformation. *Annual Review of Earth and Planetary Sciences* **28**, 169–209 (2000).
- [6] Ferretti, A., Prati, C. & Rocca, F. Permanent scatterers in SAR interferometry. *IEEE Transactions on Geoscience and Remote Sensing* **39**, 8–20 (2001).
- [7] Berardino, P., Fornaro, G., Lanari, R. & Sansosti, E. A new algorithm for surface deformation monitoring based on small baseline differential SAR interferograms. *IEEE Transactions on Geoscience and Remote Sensing* **40**, 2375–2383 (2002).
- [8] Ferretti, A. *et al.* A new algorithm for processing interferometric data-stacks: SqueeSAR. *IEEE Transactions on Geoscience and Remote Sensing* **49**, 3460–3470 (2011).
- [9] Yunjun, Z., Fattahi, H. & Amelung, F. Small baseline InSAR time series analysis: Unwrapping error correction and noise reduction. *Computers & Geosciences* **133**, 104331 (2019).
- [10] Morishita, Y. *et al.* LiCSBAS: An open-source InSAR time series analysis package integrated with the LiCSAR automated Sentinel-1 InSAR processor. *Remote Sensing* **12**, 424 (2020).
- [11] Morishita, Y. Nationwide urban ground deformation monitoring in japan using Sentinel-1 LiCSAR products and LiCSBAS. *Progress in Earth and Planetary Science* **8**, 6 (2021).

- 1059 [12] Haghshenas Haghighi, M. & Motagh, M. Uncovering the impacts of depleting aquifers:
1060 A remote sensing analysis of land subsidence in Iran. *Science Advances* **10**, eadk3039
1061 (2024).
- 1062 [13] Zebker, H. A. & Villasenor, J. Decorrelation in interferometric radar echoes. *IEEE*
1063 *Transactions on Geoscience and Remote Sensing* **30**, 950–959 (1992).
- 1064 [14] Crosetto, M., Monserrat, O., Cuevas-Gonzalez, M., Devanthery, N. & Crippa, B. Per-
1065 sistent scatterer interferometry: A review. *ISPRS Journal of Photogrammetry and*
1066 *Remote Sensing* **115**, 78–89 (2016).
- 1067 [15] Raucoules, D., Colesanti, C. & Carnec, C. Use of SAR interferometry for detecting
1068 and assessing ground subsidence. *Comptes Rendus Geoscience* **339**, 289–302 (2007).
- 1069 [16] Tomas, R. *et al.* A ground subsidence study based on DInSAR data: Calibration of
1070 soil parameters and subsidence prediction in murcia city (spain). *Engineering Geology*
1071 **111**, 19–30 (2010).
- 1072 [17] Ohenhen, L. O., Shirzaei, M., Ojha, C. & Kirwan, M. L. Hidden vulnerability of US
1073 atlantic coast to sea-level rise due to vertical land motion. *Nature Communications*
1074 **14**, 2038 (2023).
- 1075 [18] Ohenhen, L. O., Shirzaei, M., Ojha, C. *et al.* Disappearing cities on US coasts. *Nature*
1076 **627**, 108–115 (2024).
- 1077 [19] Ohenhen, L. O., Zhai, G., Shirzaei, M. *et al.* Land subsidence risk to infrastructure in
1078 US metropolises. *Nature Cities* **2**, 543–554 (2025).
- 1079 [20] Sadhasivam, N., Ohenhen, L., Khorrami, M. *et al.* Building damage risk in sinking
1080 indian megacities. *Nature Sustainability* **8**, 1467–1479 (2025).
- 1081 [21] Herrera-Garcia, G., Ezquerro, P., Tomas, R. *et al.* Mapping the global threat of land
1082 subsidence. *Science* **371**, 34–36 (2021).
- 1083 [22] Raspini, F. *et al.* Review of satellite radar interferometry for subsidence analysis.
1084 *Earth-Science Reviews* **235**, 104239 (2022).
- 1085 [23] Huning, L. S., Love, C. A., Anjileli, H. *et al.* Global land subsidence: Impact of cli-
1086 mate extremes and human activities. *Reviews of Geophysics* **62**, e2023RG000817
1087 (2024).
- 1088 [24] Gogoi, A., Kothiyari, G. C. & Patidar, A. K. Ground subsidence monitoring in india
1089 using InSAR: A review of current status and future prospects. *Geosystems and*
1090 *Geoenvironment* 100436 (2025).
- 1091 [25] Xiong, Z. *et al.* Nationwide mapping and characterization of land subsidence in the
1092 United States using InSAR. *Remote Sensing of Environment* **333**, 115134 (2026).
- 1093 [26] Yang, M., Huang, W., Li, M. & Chang, L. From PS-InSAR observations to building-
1094 level risk: A data-driven framework for urban structural resilience. *Remote Sensing of*
1095 *Environment* **341**, 115454 (2026).

- [27] Oelmann, J., Nicholls, R. J., Lincke, D. *et al.* Subsidence more than doubles sea-level rise today along densely populated coasts. *Nature Communications* **17**, 4382 (2026). 1105
- [28] Fernandez-Torres, E. A. *et al.* Country-scale assessment of urban areas, population, and households exposed to land subsidence using Sentinel-1 InSAR, and GPS time series. *Natural Hazards* **120**, 1577–1601 (2024). 1106
- [29] Ohenhen, L. O., Zhai, G., Lucy, J. *et al.* Land subsidence risk to infrastructure in US metropolises. *Nature Cities* **2**, 543–554 (2025). 1107
- [30] Luachapichatikul, S. *Measuring Land Subsidence with PS-InSAR in Bangkok using Sentinel-1 Time-Series Techniques*. Master’s thesis, Burapha University (2020). URL <https://ir.buu.ac.th/dspace/handle/1513/311>. 1108
- [31] Pumpuang, A. & Aobpaet, A. Evolution pattern of land subsidence using InSAR time-series analysis in Bangkok, Bangkok, Thailand. *Journal of Current Science and Technology* **14**, 49 (2024). URL <https://ph04.tci-thaijo.org/index.php/JCST/article/view/2549>. 1109
- [32] Aieorattanawadee, N. *Monitoring land subsidence in Bangkok during 2018–2022 using time-series InSAR analysis with MintPy*. Master’s thesis, Chulalongkorn University (2022). URL <https://digital.car.chula.ac.th/chulaetd/6585/>. 1110
- [33] Murray, K., Barbee, M., Thompson, P. & Fletcher, C. Coastal land subsidence accelerates timelines for future flood exposure in Hawai’i. *Communications Earth & Environment* **6**, 123 (2025). 1111
- [34] Blackwell, E., Shirzaei, M., Ojha, C. & Werth, S. Tracking California’s sinking coast from space: Implications for relative sea-level rise. *Science Advances* **6**, eaba4551 (2020). 1112
- [35] Goodchild, M. F. Geographical information science. *International Journal of Geographical Information Systems* **6**, 31–45 (1992). 1113
- [36] Openshaw, S. *The Modifiable Areal Unit Problem* No. 38 in Concepts and Techniques in Modern Geography (Geo Books, Norwich, 1984). 1114
- [37] Eicher, C. L. & Brewer, C. A. Dasymetric mapping and areal interpolation: Implementation and evaluation. *Cartography and Geographic Information Science* **28**, 125–138 (2001). 1115
- [38] Mennis, J. Generating surface models of population using dasymetric mapping. *The Professional Geographer* **55**, 31–42 (2003). 1116
- [39] Fotheringham, A. S. & Wong, D. W. S. The modifiable areal unit problem in multivariate statistical analysis. *Environment and Planning A* **23**, 1025–1044 (1991). 1117
- [40] Peng, R. D. Reproducible research in computational science. *Science* **334**, 1226–1227 (2011). 1118
- [41] Ohenhen, L. O. *et al.* Global subsidence of river deltas. *Nature* **649**, 894–901 (2026). 1119

- 1151 [42] Ohenhen, L. Global subsidence of river deltas. [https://doi.org/10.5281/zenodo.18092](https://doi.org/10.5281/zenodo.18092985)
1152 985 (2025). 1-km vertical land motion GeoTIFFs for 40 river deltas.
- 1153 [43] COMET. LiCSAR product details (2026). URL [https://comet.nerc.ac.uk/comet-1](https://comet.nerc.ac.uk/comet-1ics-portal-product-details/)
1154 [ics-portal-product-details/](https://comet.nerc.ac.uk/comet-1ics-portal-product-details/). Accessed 27 September 2026; product-specific coherence
1155 encoding and unwrapped validity.
- 1156 [44] Morishita, Y. LiCSBAS2: time-series processing software (2026). URL [https://github](https://github.com/yumorishita/LiCSBAS2)
1157 [.com/yumorishita/LiCSBAS2](https://github.com/yumorishita/LiCSBAS2). Commit 2627e50a1c703becd4a0b18754b05c1cf8c564d5;
1158 local Windows I/O portability patch documented.
- 1159 [45] California Department of Water Resources. Vertical displacement point data locations,
1160 2026q1: published observation-cell polygons (2026). URL [https://gis.water.ca.gov/arc](https://gis.water.ca.gov/arcgis/rest/services/Elevation/Vertical_Displacement_Point_Data_Locations_2026Q1/MapServer)
1161 [gis/rest/services/Elevation/Vertical_Displacement_Point_Data_Locations_2026Q1/Map](https://gis.water.ca.gov/arcgis/rest/services/Elevation/Vertical_Displacement_Point_Data_Locations_2026Q1/MapServer)
1162 [Server](https://gis.water.ca.gov/arcgis/rest/services/Elevation/Vertical_Displacement_Point_Data_Locations_2026Q1/MapServer). Service layer 0, geometry type polygon; accessed 27 September 2026.
- 1163 [46] California Department of Water Resources. SGMA data viewer documentation: TRE
1164 Altamira InSAR dataset (2026). URL [https://sgma.water.ca.gov/webgis/config/cus](https://sgma.water.ca.gov/webgis/config/custom/html/SGMADataViewer/doc/)
1165 [tom/html/SGMADataViewer/doc/](https://sgma.water.ca.gov/webgis/config/custom/html/SGMADataViewer/doc/). 100 m measurement support distinguished from
1166 interpolated display rasters; accessed 27 September 2026.
- 1167 [47] Sangha, S. S., Funning, G. J., Govorcin, M. & Bekaert, D. P. S. A statewide InSAR
1168 velocity map for California produced by a comprehensive large scale analysis of ARIA
1169 standard products. *Earth and Space Science* **13**, e2026EA005214 (2026).
- 1170 [48] Sangha, S., Funning, G., Govorcin, M. & Bekaert, D. Data and software for sangha et
1171 al., 2026: A statewide InSAR velocity map for California. [https://doi.org/10.5281/ze](https://doi.org/10.5281/zenodo.19493073)
1172 [nodo.19493073](https://doi.org/10.5281/zenodo.19493073) (2026). Fixed April 2026 archived version; CC BY 4.0.
- 1173 [49] City of Fresno. Public existing land use GIS layer (2026). URL [https://services2.arcg](https://services2.arcgis.com/WkBUOjyNPhsWOk1W/arcgis/rest/services/Existing_Land_Use/FeatureServer/19)
1174 [is.com/WkBUOjyNPhsWOk1W/arcgis/rest/services/Existing_Land_Use/FeatureServer](https://services2.arcgis.com/WkBUOjyNPhsWOk1W/arcgis/rest/services/Existing_Land_Use/FeatureServer/19)
1175 [/19](https://services2.arcgis.com/WkBUOjyNPhsWOk1W/arcgis/rest/services/Existing_Land_Use/FeatureServer/19). Hosted FeatureServer layer 19; ObjectID, ELU, ELUText and LayerRefreshDate;
1176 accessed 1 October 2026.
- 1177 [50] Pesaresi, M. *et al.* Advances on the global human settlement layer by joint assessment
1178 of earth observation and population survey data. *International Journal of Digital*
1179 *Earth* **17**, 2390454 (2024).
- 1180 [51] WorldPop. Thailand 2020 population counts: unconstrained, UN-adjusted, 3 arc-
1181 second product (2020). URL <https://hub.worldpop.org/geodata/summary?id=29165>.
1182 File tha_ppp_2020_UNadj.tif; accessed 27 September 2026.
- 1183 [52] Zanaga, D., Van De Kerchove, R., Daems, D. *et al.* ESA WorldCover 10 m 2021 v200.
1184 <https://doi.org/10.5281/zenodo.7254221> (2022).
- 1185 [53] Marfai, M. A. & King, L. Monitoring land subsidence in semarang, indonesia.
1186 *Environmental Geology* **53**, 651–659 (2007).
- 1187 [54] Hakim, W. L., Achmad, A. R., Eom, J. & Lee, C.-W. Land subsidence measurement
1188 of jakarta coastal area using time series interferometry with Sentinel-1 SAR data.
1189 *Journal of Coastal Research* **102**, 75–81 (2020).

- [55] Zhang, Y. *et al.* Monitoring land subsidence in wuhan city (China) using the SBAS-InSAR method with radarsat-2 imagery data. *Sensors* **19**, 743 (2019).
[56] Orhan, O., Oliver-Cabrera, T., Wdowinski, S., Yalvac, S. & Yakar, M. Land subsidence and its relations with sinkhole activity in karapinar region, turkey: a multi-sensor InSAR time series study. *Sensors* **21**, 774 (2021).
[57] Ao, Z., Hu, X., Tao, S., Hu, X. *et al.* A national-scale assessment of land subsidence in China's major cities. *Science* **384**, 301–306 (2024).
[58] Fang, J., Nicholls, R. J., Brown, S., Lincke, D. *et al.* Benefits of subsidence control for coastal flooding in China. *Nature Communications* **13**, 6946 (2022).
[59] Payne, J. A. *et al.* Widespread extent of irrecoverable aquifer depletion revealed by country-wide analysis of land surface subsidence hazard in Iran. *Journal of Geophysical Research: Solid Earth* **130**, e2024JB030367 (2025).
[60] Payne, J. Accompanying data for paper: Widespread extent of irrecoverable aquifer depletion revealed by country-wide analysis of land surface subsidence hazard in Iran. <https://doi.org/10.5281/zenodo.13754200> (2024).

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Author Contributions

Zixuan Liu: Conceptualization, Data curation, Formal analysis, Methodology, Software, Validation, Visualization, Writing—original draft, Writing—review and editing. Wei Xiong: Conceptualization, Supervision, Methodology, Validation, Writing—review and editing.

Additional Information

Competing interests

The authors declare no competing interests.

Figure legends

Figure 1. Workflow for population-weighted support accounting. The comparison fixes the geographic domain, support field and allocated total. It compares uniform coarse allocation with preservation of the native population–support cross moment. Neither branch infers motion in unsupported locations.

Figure 2. Chao Phraya population and built-surface support deficits in the fixed geographic domain at $q = 0.3$. Red bars show uniform allocation within 1-km cells averaged across four origins; narrow whiskers span those origins. Blue bars show the native-model reference. These are conditional model-allocation quantities, not unique residents or confirmed subsidence exposure. The submitted 3.77-million statistic uses a different domain and binary selection; its lineage is preserved in Supplementary Section S2.

Figure 3. Continuous support sensitivity. Curves give the fraction of allocated population or built-surface quantity with native-pixel mean support below t , for joint unwrapped/coherence cutoffs $q = 0.2, 0.3, 0.4$. The domain and allocated totals are identical across cutoffs. The display uses 101 threshold values; the exact empirical-distribution integral is checked separately. No velocity threshold determines these curves.

Figure 4. Pair-sampling sensitivity. (a) Means and central 95% ranges from 100 draws at each size within the fixed 224-pair pool, at $q = 0.3$. At 224 pairs the whole pool is selected and the range collapses. (b) Population and built-surface deficit shares in the fixed nested selections. The nested 28 pairs differ from the legacy 28-pair sample. Ranges condition on the archived pool; neither panel supplies calibrated uncertainty for household population or true deformation.

Figure 5. Regional transfer under specified support fields. (a) Uniform-minus-native population support deficits for 1-km cells, in percentage points of each fixed AOI total; whiskers span four grid origins. (b) Native population-weighted support in the same domains. Chao Phraya uses sampled mean area-pair support at $q = 0.3$; Fresno uses the nonduplicated DWR observation-polygon union; Los Angeles and Davis-Sacramento use their documented ARIA quality-support definitions at the displayed 0.3 cutoff. The support objects differ, and the effects are not pooled or interpreted as a universal correction factor.

Figure 6. External GNSS consistency checks. Native-pixel California InSAR-GNSS LOS-rate RMSE is shown for the archived provider frame and the location-selected local frame. Control sites are excluded from the 11 Los Angeles and four Davis-Sacramento validation stations. Both reference choices are retained. Temporal-model, source-version and station-dependence limitations are detailed in Supplementary Section S30. These comparisons do not validate the separate Bangkok inversion or genuinely missing deformation.

Table legends

Table 1. Native-model support deficits in the primary geographic domain and legacy 28-pair sample. Each percentage uses its stated population or area denominator.

Table 2. Checks on support accounting, grouped by the alternative explanation or information limitation they address. The table summarizes experiments detailed in the main Results and Supplementary Sections S4-S30; it is not a combined validation score.